



Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LESSON PLAN

For

Certified Lean Six Sigma White Belt (CLSSWB) Training

Course Code: C142

Conducted by

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

Version 3

DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
1	1 July 2026	Initial release — CLSSWB 1-day lesson plan for the Lean Six Sigma awareness course.	Dr. Alfred Ang
2	19 July 2026	Rebuilt from the single-source content module: DMAIC roadmap end to end within one 8-hour training day, exactly one hands-on lab per DMAIC phase (5 labs) on a single continuous scenario, content simplified to White Belt awareness depth.	Dr. Alfred Ang
3	21 July 2026	Released under course code C142 as a commercial one-day hands-on course. Schedule restructured so every DMAIC lab gains additional hands-on time and the day closes with a full course recap and Q&A.	Dr. Alfred Ang

TABLE OF CONTENTS

Course Information.....	4
Learning Outcomes.....	4
Learning Reinforcement.....	4
Course Schedule.....	4
Day 1 — Lean Six Sigma foundations and the full DMAIC roadmap, hands-on.....	4
Lab Reference (aligned to course topics).....	5

Course Information

Course Title	Certified Lean Six Sigma White Belt (CLSSWB) Training
Course Reference	C142
Training Provider	Tertiary Infotech Academy Pte Ltd (UEN 201200696W)
Duration	1 day · 8 training hours per day (8 hours)
Daily Timing	9:30 am – 6:30 pm (1-hour lunch; tea breaks within training time)
Mode	Instructor-led, hands-on Lean Six Sigma labs using the BrewBean Cafe morning rush improvement scenario
Trainer	Dr. Alfred Ang

Learning Outcomes

On completion of this course, learners will be able to:

- LO1: Describe the core concepts of quality, Lean, Six Sigma and Lean Six Sigma, and the belt roles in an improvement team.
- LO2: Explain the five phases of the DMAIC roadmap and what happens in each.
- LO3: Identify customer requirements and a clear problem statement in the Define phase.
- LO4: Describe how a process is mapped and measured, and identify the eight wastes in the Measure phase.
- LO5: Identify likely causes of a problem using 5 Whys and Fishbone analysis in the Analyze phase.
- LO6: Describe common improvement and control actions that fix a cause and hold the gain.

Learning Reinforcement

- Every topic is followed by hands-on labs that learners complete on their own machine.
- Each lab ends with a 'Test it' verification step so learners can confirm their own progress.
- The trainer circulates during lab time to give individual feedback and unblock learners.
- Each topic closes with a recap mapped back to the course learning outcomes.

Course Schedule

Day 1 — Lean Six Sigma foundations and the full DMAIC roadmap, hands-on

Time	Duration	Topic / Activity
9:30–10:00	30 min	Welcome, course introduction and ground rules
10:00–11:00	60 min	FOUNDATIONS — What is Quality; What is Lean; What is Six Sigma; Lean vs Six Sigma vs Lean Six Sigma;

		the belt roles and where a White Belt contributes; the DMAIC roadmap
11:00–11:15	15 min	Tea break
11:15–12:00	45 min	DMAIC · DEFINE — Voice of the Customer; VOC to CTQ translation; problem statements; SMART goals; the project charter and scope
12:00–12:40	40 min	Hands-on: Lab 1: Define — Customer Requirements and the Problem Statement
12:40–13:40	60 min	Lunch break
13:40–14:25	45 min	DMAIC · MEASURE — process mapping; SIPOC; types of data; data collection plans; check sheets; the eight wastes (DOWNTIME); value-added analysis
14:25–15:05	40 min	Hands-on: Lab 2: Measure — Map the Process and Spot the Waste
15:05–15:50	45 min	DMAIC · ANALYZE — symptom vs root cause; 5 Whys; Fishbone (5M); reading a Pareto chart; common vs special cause variation
15:50–16:05	15 min	Tea break
16:05–16:45	40 min	Hands-on: Lab 3: Analyze — Find the Root Cause using the 5 Whys and Fishbone tools
16:45–17:10	25 min	DMAIC · IMPROVE — generating solutions; impact/effort screening; 5S; mistake proofing (Poka-Yoke); standard work; piloting
17:10–17:35	25 min	Hands-on: Lab 4: Improve — Choose and Pilot a Countermeasure
17:35–17:55	20 min	DMAIC · CONTROL — the control plan; visual management; SOPs; team huddles; handover
17:55–18:15	20 min	Hands-on: Lab 5: Control — Hold the Gain and Hand Over
18:15–18:30	15 min	Course recap and Q&A

Total training time: 480 minutes (8 hours).

Lab Reference (aligned to course topics)

Topic	Coverage	Labs
Topic 01: Lean Six Sigma Foundations	20%	
Topic 02: Define — Understand the Problem	20%	Lab 1

Topic 03: Measure — See What Is Really Happening	20%	Lab 2
Topic 04: Analyze — Find the Cause	20%	Lab 3
Topic 05: Improve — Fix the Cause	10%	Lab 4
Topic 06: Control — Hold the Gain	10%	Lab 5